

LAKSHYA

JEE 2027

Lecture No.- 01

MATHEMATICS

**METHOD OF
DIFFERENTIATION(M.O.D.)**

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Topics *to be covered*



1 Finding Functions

2 Rules & Formula's of Differentiation

3 Question Practice

4



H.W. Discussion

QUESTION



Let a and c be real numbers such that $c > 0$ and $f(x)$ is defined on $[0, 1]$ as

$$f(x) = \begin{cases} x^a \sin(x^{-c}), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

Show that

- (i) $f(x)$ is continuous if and only if $a > 0$
- (ii) $f'(0)$ exists if and only if $a > 1$
- (iii) $f'(x)$ is continuous if and only if $a > 1 + c$
- (iv) $f''(0)$ exists if and only if $a > 2 + c$

$f'(0)$ exist
↳ $f(x)$ is diff. at $x=0$

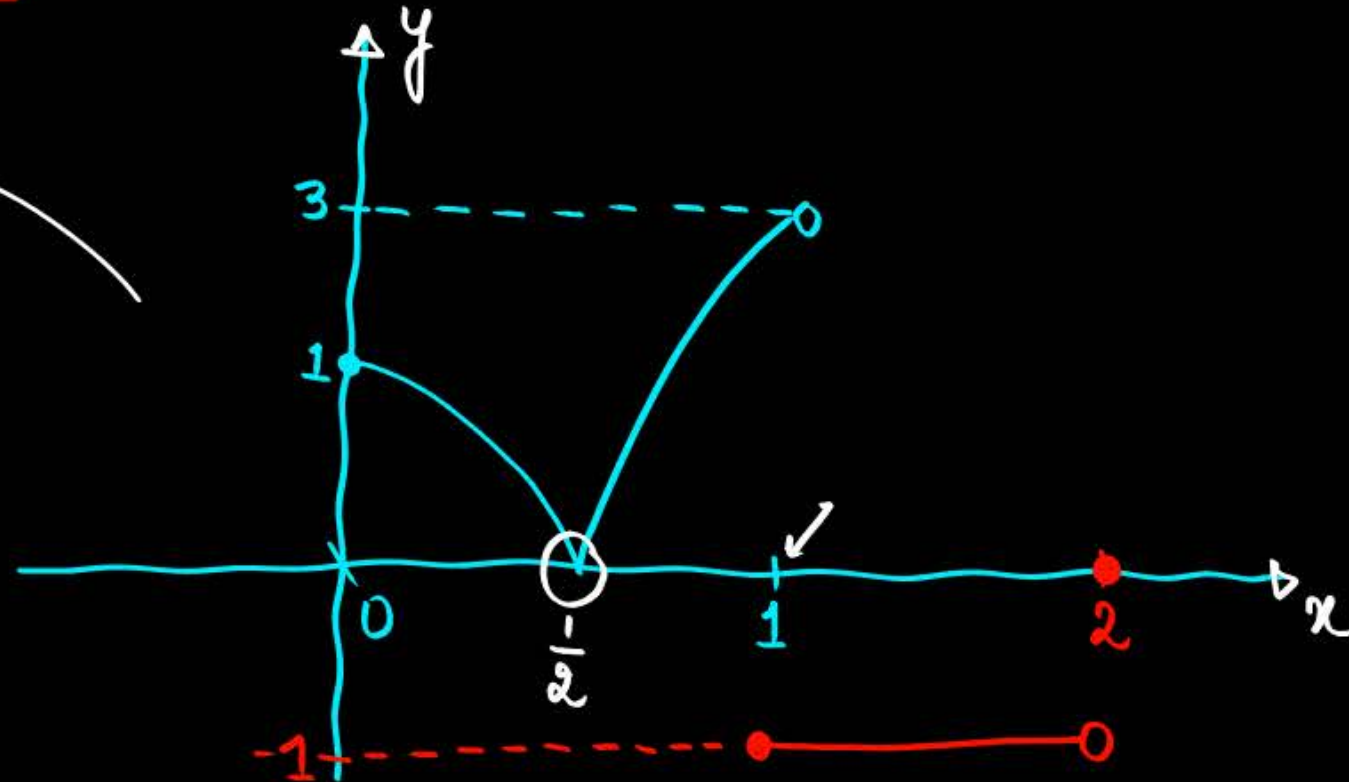
$f''(0)$ exist
↳ $f'(x)$ is diff. at $x=0$

$$f(x) = \begin{cases} |1 - 4x^2|, & 0 \leq x < 1 \\ [x^2 - 2x], & 1 \leq x \leq 2 \end{cases}$$

Check the differentiability in $(0, 2)$ where $[]$ denotes greatest integer function.

$\text{NOPND} = 2$

$x = \frac{1}{2}, 1$



Question [JEE (Adv.)–2020, Paper–1]



Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^3 - x^2 + (x - 1)\sin x$ and let $g: \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function. Let $fg: \mathbb{R} \rightarrow \mathbb{R}$ be the product function defined by $(fg)(x) = f(x)g(x)$. Then which of the following statements is/are TRUE?

- ☒ A If g is continuous at $x = 1$, then fg is differentiable at $x = 1$ (R-1)
- ☐ B If fg is differentiable at $x = 1$, then g is continuous at $x = 1$
- ☒ C If g is differentiable at $x = 1$, then fg is differentiable at $x = 1$ (R-1)
- ☐ D If fg is differentiable at $x = 1$, then g is differentiable at $x = 1$

Question [JEE (Adv.)-2020, Paper-1]



Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^3 - x^2 + (x-1)\sin x$ and let $g: \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function. Let $fg: \mathbb{R} \rightarrow \mathbb{R}$ be the product function defined by $(fg)(x) = f(x)g(x)$. Then which of the following statements is/are TRUE?

$$h(x) = f(x) \cdot g(x)$$

diff. ✓
(value = 0
at $x=1$)

at $x=1$

$$RHD = \lim_{h \rightarrow 0^+} \frac{f(1+h)g(1+h) - f(1)g(1)}{h}$$

$$LHD = \lim_{h \rightarrow 0^+} \frac{f(1-h)g(1-h) - f(1)g(1)}{-h}$$

equal
given
in opt-B, D.

diff. always

$$\# f(1) = 0$$

ex:-
 $g(1^+) = g(1^-) = 2$
 $g(1) = 3$

$$RHD = LHD$$

at $x=1$

$$RHD = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} = \frac{f(1+h)}{h}$$

$$LHD = \lim_{h \rightarrow 0^+} \frac{f(1-h) - f(1)}{-h} = \frac{f(1-h)}{-h}$$

Question [JEE-1999, 2M]



The function $f(x) = (x^2 - 1)|x^2 - 3x + 2| + \cos(|x|)$ is not differentiable at $x = \underline{\hspace{1cm}}$

A -1

B 0

C 1

☒ D 2

$$f(x) = \boxed{(\underline{x-1})(x+1) \cdot |(\underline{x-1})(x-2)|} + \underbrace{\cos x}_{\substack{\text{dummy mod.} \\ \text{diff. always } \checkmark}}$$

S.P. $\Rightarrow x = 1, 2$

$\underline{x=1}$
 $\downarrow$
Repeated root
 $\downarrow$
diff. $\checkmark$

$$\underline{x=2}$$
$$\boxed{(x-1)(x+1) \cdot |x-1|} \cdot \boxed{|x-2|}$$

diff. at $x=2$ value $\neq 0$

cont. value = 0

} (R-4) $\rightarrow x=2 \rightarrow \underline{\underline{n.d.}}$

Question [JEE (Adv.)-2016, Paper-1]

Let $a, b \in \mathbb{R}$ and $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = a \cos(|x^3 - x|) + b|x|\sin(|x^3 + x|)$. Then f is

$$\begin{aligned} &|x| \sin(|x(x^2+1)|) \\ &\underline{\underline{|x|}} \sin(\underline{\underline{|x|(x^2+1)}}) \end{aligned}$$

☒ **A** Differentiable at $x = 0$ if $a = 0$ and $b = 1$

☒ **B** Differentiable at $x = 1$ if $a = 1$ and $b = 0$

☐ **C** NOT differentiable at $x = 0$ if $a = 1$ and $b = 0$

☐ **D** NOT differentiable at $x = 1$ if $a = 1$ and $b = 1$

dummy mod sign.

$$f(x) = a \cos(x^3 - x) + b x \sin(x^3 + x)$$

diff. always

Question [JEE Mains-2019]



RAM

New to Me!!



Let f be a differentiable function from $\mathbb{R}$ to $\mathbb{R}$ such that

$|f(x) - f(y)| \leq 2|x - y|^{3/2}$, for all $x, y \in \mathbb{R}$. If $f(0) = 1$, then $f(2)$ equal to

~~A~~ 1

B 0

C $1/2$

D 2

$$|f(x) - f(y)| \leq 2|x - y| \cdot |x - y|^{1/2}$$

$$\lim_{y \rightarrow x} \frac{|f(x) - f(y)|}{|x - y|} \leq \lim_{y \rightarrow x} 2|x - y|^{1/2}$$

$$|f'(x)| \leq 0 \Rightarrow f'(x) = 0$$

$\hookrightarrow f(x)$ is a const. func?

$$\lim_{y \rightarrow x} \left| \frac{f(x) - f(y)}{x - y} \right|$$
$$\left| \lim_{y \rightarrow x} \left(\frac{f(x) - f(y)}{x - y} \right) \right|$$

$$\left| \lim_{y \rightarrow x} \left(\frac{f(x) - f(y)}{x - y} \right) \right| \stackrel{\frac{0}{0}}{=} |f'(x)|$$

$\xrightarrow{\text{d-H Rule}}$

$$\lim_{y \rightarrow x} \frac{0 + f'(y)}{0 + 1} = f'(x)$$

$\swarrow$ var. $\searrow$ const.



FINDING FUNCTIONS



Solving funcⁿ equation



Solving Function equations :

Step-1 : Write $f'(x) = \lim_{h \rightarrow 0^+} \left(\frac{f(x+h) - f(x)}{h} \right)$

Step-2 : Simplify above relation using given function equation.

Step-3 : Put $f'(x) = \frac{dy}{dx}$, $f(x) = y \rightarrow$ form a **D.E.**

Step-4 : Integrate & obtain y or $f(x)$ in terms of x .

Step-5 : Use given initial condition to find constants of integration.

Question

①

$$x=y=0 \Rightarrow f(0) = (f(0))^2 \Rightarrow f(0) = \frac{0}{x}, 1 \checkmark$$



If $f(x+y) = f(x) \cdot f(y) \forall x, y \in \mathbb{R}$ and $f(0) \neq 0$ and $f(x)$ is a differentiable function, then find $f(x)$.

$$f'(x) = \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

★★★★★★★★★★★★
(x as constant & h as variable)

$$f'(x) = \lim_{h \rightarrow 0^+} \frac{f(x) \cdot f(h) - f(x)}{h}$$

$$\# f'(x) = f(x) \left(\lim_{h \rightarrow 0^+} \frac{f(h) - 1}{h} \right)$$

$f'(0)$

$$f'(x) = f(x) \cdot f'(0)$$

const. = k

$$f'(x) = k \cdot f(x)$$

$$\frac{dy}{dx} = k \cdot y \quad (\text{Q.E.})$$

$$\# \frac{dy}{dx} = ky$$

$$\int \frac{dy}{y} = k \int dx$$

$$\ln y = kx + C$$

$$y = e^{kx+C} = f(x), \quad f(0) = 1.$$

$$x=0$$

$$e^C = 1$$

$$\hookrightarrow C=0$$

$$f(x) = e^{kx}$$

Question



If $f(x + y) = f(x) \cdot f(y) \forall x, y \in \mathbb{R}$ and $f(0) \neq 0$ and $f(x)$ is a differentiable function, then find $f(x)$.

M-2 :- Shortcut :-

diff. w.r.to y keeping x const. & then put $y=0$.

$$f'(x+y) \times (0+1) = f(x) \cdot f'(y)$$

$\hookrightarrow y=0$

$$f'(x) = f(x) \cdot \overset{k}{f'(0)}$$

Suppose a differentiable function $f(x)$ satisfies the identity

$f(x + y) = f(x) + f(y) + xy^2 + x^2y$, for all real x and y . If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$, then $f'(3) + f(3)$ is equal to _____.

10.

M-I:

$$f'(x) = \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0^+} \frac{\cancel{f(x)} + f(h) + xh^2 + x^2h - \cancel{f(x)}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0^+} \left(\boxed{\frac{f(h)}{h}} + xh + x^2 \right)$$

$$f'(x) = 1 + x^2$$

$$\frac{dy}{dx} = 1 + x^2$$

$$\int dy = \int (1 + x^2) dx$$

$$y = \boxed{x + \frac{x^3}{3} + C = f(x)}$$

$$x = 3$$

$$f'(3) = 10$$

Question [JEE Mains-2020]

2



Suppose a differentiable function $f(x)$ satisfies the identity

$f(x + y) = f(x) + f(y) + xy^2 + x^2y$, for all real x and y . If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$, then

$f'(3) + f(3)$ is equal to 22.

$f'(3)$ is 10.
 $f(3)$ is 12.

$$f(x) = x + \frac{x^3}{3} + C$$

$x=0$

$$f(0) = C = 0$$

$x=3$

$$f(3) = 3 + 9 = 12$$

$$x=y=0 \Rightarrow f(0) = f(0) + f(0) + 0 + 0$$

$$f(0) = 2f(0)$$

$$\# f(0) = 0$$

Question [JEE Mains-2020]

2



Suppose a differentiable function $f(x)$ satisfies the identity

$f(x + y) = f(x) + f(y) + xy^2 + x^2y$, for all real x and y . If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$, then $f'(3) + f(3)$ is equal to _____.

$\frac{0}{0}$

$$\frac{f(x)}{x}$$

L-H Rule.

$$\lim_{x \rightarrow 0} \frac{f'(x)}{1} = f'(0) = 1$$

M-2 :

$$f'(x+y) = 0 + f'(y) + x(2y) + x^2(1)$$

$y=0$

$$f'(x) = f'(0) + x^2$$

$$f'(x) = 1 + x^2$$

Question



Given a function g which has derivative $g'(x)$ for all x satisfies $g'(0) = 2$ and $g(x + y) = e^y g(x) + e^x g(y)$ for all $x, y \in \mathbb{R}$, $g(5) = 32$. Find the value of $g'(5) - 2e^5$.



Question

$$\# f(x) = 2x + 1$$



Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a real-valued differentiable function satisfying

$$f\left(\frac{2x+3y}{5}\right) = \frac{2f(x)+3f(y)}{5} \text{ for all } x, y \in \mathbb{R} \text{ and } f(0) = 1, f'(0) = 2. \text{ Find the value}$$

of $f(3) + f'(4)$

$$\downarrow \quad \downarrow$$

$$7 + 2 = 9$$

Ans

diff. w.r. to y keeping x const

$$f'\left(\frac{2x+3y}{5}\right) \times \frac{3}{5} = \frac{1}{5} (0 + 3f'(y))$$

$$y=0$$

$$f'\left(\frac{2x}{5}\right) \times 3 = 3f'(0)$$

$$f'\left(\frac{2x}{5}\right) = 2$$

$$x \rightarrow \frac{5}{2}x$$

$$f'(x) = 2$$

$$x=0, y=1$$

$$\frac{dy}{dx} = 2$$

$$1 = 0 + C$$

$$1 = C$$

$$\int dy = \int 2 dx$$

$$y = 2x + C$$

Question



A function $f: (0, \infty) \rightarrow \mathbb{R}$ satisfies the equation $f(x/y) = f(x) - f(y)$. If $f(x)$ is differentiable on $(0, \infty)$ and $\lim_{x \rightarrow 0} \frac{f(1+x)}{x} = 3$, then determine $f(x)$.

Ans

Paragraph for Question Nos. A to C

Let f be a differentiable function satisfying the relation

$$f(x+y) = f(x) + f(y) - 2xy + (e^x - 1)(e^y - 1) \quad \forall x, y \in \mathbb{R} \text{ and } f'(0) = 1.$$

Question [A]

$(f(2))$ is equal to

[Note: $\{y\}$ denotes the fraction part function of y .]

A $e^2 - 5$

B $e^2 - 6$

C $e^2 - 7$

D $e^2 - 8$

$f(2)$

Paragraph for Question Nos. A to C

Let f be a differentiable function satisfying the relation

$$f(x+y) = f(x) + f(y) - 2xy + (e^x - 1)(e^y - 1) \quad \forall x, y \in \mathbb{R} \text{ and } f'(0) = 1.$$

Question [B]

The value of $\lim_{x \rightarrow 0} \frac{f(2x) + 4x^2 - 2x}{x^2}$ is

A 2

B 4

C 8

D 16

Ans.

Paragraph for Question Nos. A to C

Let f be a differentiable function satisfying the relation

$$f(x+y) = f(x) + f(y) - 2xy + (e^x - 1)(e^y - 1) \quad \forall x, y \in R \text{ and } f'(0) = 1.$$

Question [C]

Let $g(x) = f(x) + x^2 - 2$. If the equation $|g(|x|)| = k$ has four distinct solutions, then the set of values of k is

Two

- A $(0, 1)$
- B $(-2, 2)$
- C $(0, 3)$
- D $(0, 2)$

Question [IIT-JEE-2011, Paper-1]

Ans. : (B,C)



Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x+y) = f(x) + f(y), \forall x, y \in \mathbb{R}$. If $f(x)$ is differentiable at $x = 0$, then

- A** $f(x)$ is differentiable only in a finite interval containing zero
- B** $f(x)$ is continuous $\forall x \in \mathbb{R}$
- C** $f'(x)$ is constant $\forall x \in \mathbb{R}$
- D** $f(x)$ is differentiable except at finitely many points

QIBY!!

METHOD OF DIFFERENTIATION

- easy
- formula based
- always used
- Mains - 1 Quesⁿ



INTRODUCTION

If $y = f(x)$, then derivative of y with respect to x is given by,

$$f'(x) = \frac{dy}{dx} = \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

This is also known as **differential coefficient.**

AB - Initio Method OR First principle (Class - XI)

$$f'(x) = \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

If $f(x) = x^5$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0^+} \frac{(x+h)^5 - x^5}{h} = \lim_{h \rightarrow 0^+} \frac{x^5 \left(\left(1 + \frac{h}{x}\right)^5 - 1 \right)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{x^5 \left(1 + 5 \frac{h}{x} + \dots - 1 \right)}{h} \\ &= x^5 \frac{5}{x} \\ &= 5x^4. \end{aligned}$$



BASIC FORMULA



$$(1) \quad \frac{d}{dx}(\text{constant}) = 0$$

$$(2) \quad \frac{d}{dx}(x^n) = n \cdot x^{n-1}$$

$$(3) \quad \frac{d}{dx}(e^x) = e^x$$

$$(4) \quad \frac{d}{dx}(a^x) = a^x \cdot \ln a$$

$$(5) \quad \frac{d}{dx}(\ln x) = \frac{1}{x} \text{ or } \frac{d}{dx}(\ln|x|) = \frac{1}{x}$$

$$(6) \quad \frac{d}{dx}(\log_a x) = \frac{1}{x} \log_a e = \frac{1}{x \ln a}$$

$$(7) \quad \frac{d}{dx}(\sin x) = \cos x$$

$$(8) \quad \frac{d}{dx}(\underline{\cos} x) = \ominus \sin x$$

$$(9) \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

$$(10) \quad \frac{d}{dx}(\underline{\cot} x) = \ominus \operatorname{cosec}^2 x$$

$$(11) \quad \frac{d}{dx}(\sec x) = \sec x \cdot \tan x$$

$$(12) \quad \frac{d}{dx}(\underline{\operatorname{cosec}} x) = \ominus \operatorname{cosec} x \cdot \cot x$$



DERIVATIVES OF I.T.F.



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$$(13) \quad \frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}; |x| \leq 1$$

$$(14) \quad \frac{d}{dx} (\underline{\underline{\cos}}^{-1} x) = \frac{-1}{\sqrt{1-x^2}}; |x| \leq 1$$

$$(15) \quad \frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}; x \in \mathbb{R}$$

$$(16) \quad \frac{d}{dx} (\underline{\underline{\cot}}^{-1} x) = \frac{-1}{1+x^2}; x \in \mathbb{R}$$

$$(17) \quad \frac{d}{dx} (\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}; |x| \geq 1$$

$$(18) \quad \frac{d}{dx} (\underline{\underline{\csc}}^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}; |x| \geq 1$$

Proof:-

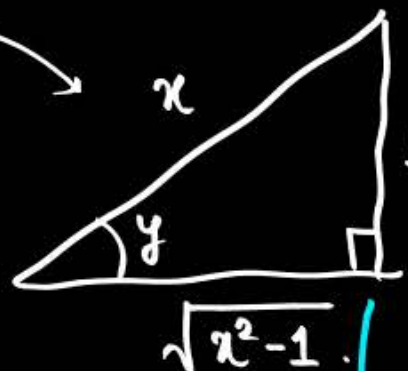
$y = \operatorname{cosec}^{-1} x$, $x \in (-\infty, -1] \cup [1, \infty)$

$y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

$\operatorname{cosec} y = \frac{x}{1}$
diff. w.r. to x :

$-\operatorname{cosec} y \cdot \cot y \cdot \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{-1}{\operatorname{cosec} y \cdot \cot y}$



$y \in \text{I quad.}$

$\cot y = +\sqrt{x^2 - 1}$

$y \in \text{IV quad.}$

$\cot y = -\sqrt{x^2 - 1}$

$\frac{dy}{dx} = \frac{-1}{\operatorname{cosec} y \cdot \cot y}$
 (Note: cosec y is circled in green with an arrow pointing to x)

$\frac{dy}{dx} = \frac{-1}{x \cdot \sqrt{x^2 - 1}}$, $x \geq 1$

$\frac{dy}{dx} = \frac{+1}{x \cdot (+\sqrt{x^2 - 1})}$, $x \leq -1$

$\frac{dy}{dx} = \frac{-1}{|x| \cdot \sqrt{x^2 - 1}}$

$$(i) \quad \frac{d}{dx} (\sqrt{f(x)}) = \frac{1}{2\sqrt{f(x)}} \cdot f'(x)$$

$$(ii) \quad \frac{d}{dx} \left(\frac{1}{f(x)} \right) = \frac{-1}{(f(x))^2} \cdot f'(x)$$

$$(iii) \quad \frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

ex:- $\sqrt{x^3+3x} \xrightarrow{d} \frac{1}{2\sqrt{x^3+3x}} \times (3x^2+3)$

ex:- $\frac{1}{\sin x + 3 \cos x} \xrightarrow{d} \frac{-1}{(\sin x + 3 \cos x)^2} \times (\cos x - 3 \sin x)$



RULES OF DIFFERENTIATION

1. If $y = f(x) \pm u(x) \pm h(x)$

$$\Rightarrow \frac{dy}{dx} = f'(x) \pm u'(x) \pm h'(x)$$

2. If $y = k \cdot f(x)$

$$\Rightarrow \frac{dy}{dx} = k f'(x)$$



RULES OF DIFFERENTIATION

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PRODUCT RULE

$$\Rightarrow \text{If } y = u \cdot v \Rightarrow \frac{dy}{dx} = u \cdot \frac{dv}{dx} + v \cdot \frac{du}{dx}$$

Proof:-

$$\text{Let } h(x) = f(x) \cdot g(x)$$

$$h'(x) = \lim_{h \rightarrow 0^+} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x)}{h}$$

$$\lim_{h \rightarrow 0^+} \left(f(x+h) \cdot \left(\frac{g(x+h) - g(x)}{h} \right) + g(x) \cdot \left(\frac{f(x+h) - f(x)}{h} \right) \right)$$
$$= \underbrace{f(x+h) \cdot g(x+h)}_h - \underbrace{f(x+h) \cdot g(x)}_h + \underbrace{f(x+h) \cdot g'(x)}_h - \underbrace{f(x) \cdot g'(x)}_h$$

$$h'(x) = f(x) \cdot g'(x) + g(x) \cdot f'(x)$$



RULES OF DIFFERENTIATION

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QUOTIENT RULE

$$\Rightarrow \text{If } y = \frac{u}{v} \Rightarrow \frac{dy}{dx} = \left(\frac{v \cdot \frac{du}{dx} - u \cdot \frac{dv}{dx}}{v^2} \right)$$

Ex.: If $y = \frac{x \sin x}{\ln x}$, then find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{(x \cos x + \sin x) \ln x - (x \sin x) \frac{1}{x}}{(\ln x)^2}$$

Proof:-

$$y = \frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)}$$
$$\frac{dy}{dx} = f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \frac{-1}{(g(x))^2} \cdot g'(x)$$
$$= \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$$



RULES OF DIFFERENTIATION

CHAIN RULE (Derivative of Composite Function)

$$\Rightarrow \text{If } y = f(g(h(x))) \Rightarrow \frac{dy}{dx} = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x) \cdot 1.$$

Ex.: If $y = \ln(\tan(\sqrt{1+x^2}))$ then find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{1}{\tan \sqrt{1+x^2}} \times \sec^2(\sqrt{1+x^2}) \times \frac{1}{2\sqrt{1+x^2}} \times 2x.$$



(i) $D\left(\frac{1}{\sin x}\right) = \frac{-1}{(\sin x)^2} \times \cos x$

(ii) $D\left(\tan(\tan^{-1} x)\right) = D(x) = 1$

(iii) $D\left(\cos^4 \frac{x}{2} - \sin^4 \frac{x}{2}\right) = D(\cos x) = -\sin x$

(iv) $D(\cos^{-1} x + \sin^{-1} x)^n = D\left(\frac{\pi}{2}\right)^n = 0$

(v) $D(\cos x \operatorname{cosec} x) = D(\cot x) = -\operatorname{cosec}^2 x$

(vi) $D\left(\frac{1}{\log_x e}\right)$ when $x = \frac{1}{e} \Rightarrow D(\ln x) = \frac{1}{x} \Big|_{x=\frac{1}{e}} = e$

(vii) $D\left(\frac{1-\cos 2x}{\sin 2x}\right) = D(\tan x) = \sec^2 x$

$$\begin{aligned} & \cos^4 \frac{x}{2} - \sin^4 \frac{x}{2} \\ & \left(\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} \right) \times \left(\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2} \right) \\ & \quad \downarrow \qquad \qquad \downarrow \\ & \cos x \qquad \qquad 1 \end{aligned}$$



$$(ix) \quad D([x]) = 0 \quad \forall x \in \mathbb{R} - \mathbb{Z}.$$

$$(x) \quad D(\{x\}) = 1 \quad \forall x \in \mathbb{R} - \mathbb{Z}.$$

$$(xi) \quad D(\tan^{-1}(\tan x)) = 1 \quad \forall x \in \mathbb{R} - \left\{(2n+1)\frac{\pi}{2}\right\}$$

$$(xii) \quad D(|x|) = \begin{cases} x > 0 & \rightarrow D(x) = 1 \\ x < 0 & \rightarrow D(-x) = -1 \end{cases}$$

$$(xiii) \quad D(\text{sgn}(x)) = 0 \quad \forall x \in \mathbb{R} - \{0\}$$

Question [JEE Mains-2024]



If $f(x) = ax^3 + bx^2 + cx + 41$ be such that $f(1) = 40$, $f'(1) = 2$ and $f''(1) = 4$. then $a^2 + b^2 + c^2$ is equal to

51 fnf.

$$x=1 \Rightarrow f(1) = a + b + c + 41 = 40.$$

diff. w.r. to x :

$$f'(x) = 3ax^2 + 2bx + c$$

$x=1$

$$f'(1) = 3a + 2b + c = 2$$

diff. w.r. to x :

$$f''(x) = 6ax + 2b$$

$x=1$

$$6a + 2b = 4.$$

Acidic Question [JEE MAINS]



Let $f(x) = x^3 + x^2 f'(1) + 2x f''(2) + f'''(3)$, $x \in R$. Then the value of $f'(5)$ is

a

b

c

$x=5$

$$f'(5) = 75 + 10a + 2b$$

$$f(x) = x^3 + ax^2 + 2bx + c$$

$x=1$

$$f'(1) = 3 + 2a + 2b = a$$

$$f'(x) = 3x^2 + 2ax + 2b$$

$x=2$

$$f''(2) = 12 + 2a = b$$

$$f''(x) = 6x + 2a$$

$x=3$

$$f'''(3) = 6 = c$$

$$f'''(x) = 6$$

$$a + 2b = -3$$

$$a + 2(12 + 2a) = -3$$

$$5a = -27$$

$$a = -\frac{27}{5}$$

Question [JEE Mains-2024]

[Ans. 14]



If $f(x) = x^2 + g'(1)x + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$, the value of $f(4) - g(4)$ is equal to _____.

Ans.

Question [JEE Mains-2023]

[Ans. C]



If $f(x) = x^3 - x^2 f'(1) + x f''(2) - f'''(3)$, $x \in R$, then:

- A** $3f(1) + f(2) = f(3)$
- B** $f(3) - f(2) = f(1)$
- C** $2f(0) - f(1) + f(3) = f(2)$
- D** $f(1) + f(2) + f(3) = f(0)$

QIBY.



Check differentiability of following :

(i) $f(x) = \begin{cases} 3^x, & x \in [-1, 1] \\ 4 - x, & x \in [1, 4] \end{cases}$ **at $x = 1$.**

(ii) $f(x) = x \left(\frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}} \right), x \neq 0$ & $f(0) = 0$, **at $x = 0$.**

DIBY-A

Discuss differentiability of function

$$f(x) = \begin{cases} 2x - 2, & x < 1 \\ x^2, & 1 \leq x \leq 2 \\ -2 + 8x - x^2, & x > 2 \end{cases} \text{ at } x = 1 \text{ and } x = 2$$

DIBY-B

If $f(x) = \begin{cases} x^3 + 2x^2 + ax + 6, & x < 1 \\ 2x + b, & x \geq 1 \end{cases}$ is differentiable for all $x \in R$, then find the value of $(a + b)$

DIBY-C

Let $f(x) = |x - 2|e^x g(x) + \sin x$, where $g(x)$ is a continuous function. If $f(x)$ is differentiable at $x = 2$, then find $g(2)$

DIBY-D

Let $f(x) = \begin{cases} x^2 \left| \cos \frac{\pi}{2x} \right|, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Find $f' \left(\frac{1}{3}^+ \right) + f' \left(\frac{1}{3}^- \right)$.

DIBY-E

$f(x) = \lim_{x \rightarrow \infty} \frac{x^n - \sin n x^n}{x^n + \sin n x^n} x > 0; x \neq 1, f(1) = 0$ discuss differentiability at $x = 1$

DIBY-F

$f(x) = \begin{cases} \left\{ x + \frac{1}{3} \right\} [\sin \pi x] & 0 \leq x < 1 \\ [2x] \operatorname{sgn}(x - 4/3) & 1 \leq x \leq 2 \end{cases}$ check differentiability at $x = 1$ where $[\cdot]$ is GIF
 $\{\cdot\}$ is FPF.

DIBY-G

$$f(x) = \begin{cases} x^2 \operatorname{sgn}[x] + \{x\} & 0 \leq x < 2 \\ \sin x + |x - 3| & 2 \leq x \leq 4 \end{cases} \text{ . Check differentiability at } x = 2.$$

where $[]$ is GIF & $\{ \}$ is FPF

DIBY-H

$$\text{Let } f(x) = \begin{cases} \frac{5e^{\frac{1}{x}} + 2}{3 - e^{\frac{1}{x}}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Check (i) $f(x)$ (ii) $x^2 f(x)$ for continuity and differentiability at $x = 0$.

DIBY-I

Let a_1 and a_2 be two values of a for which

$$f(x) = \begin{cases} x \cdot \left(\frac{\ln(1+x) + \ln(1-x)}{\sec x - \cos x} \right), & x \in (-1, 0) \\ (a^2 - 3a - 1)\sin x + x^2, & x \in [0, \infty) \end{cases} \text{ is differentiable at } x = 0, \text{ then find the value of } (a_1^2 + a_2^2).$$

DIBY-J

Let $f(x) = \begin{cases} x \tan^{-1} x + \sec^{-1} \frac{1}{x}, & x \in (-1, 1) - \{0\} \\ \frac{\pi}{2}, & x = 0 \end{cases}$ If $l = f'(0^+)$ and $m = f'(0^-)$, then find the value of $l^2 + m^2$.

DIBY-K

Let $g(x) = \begin{cases} \frac{x^2 + x \tan x - x \tan 2x}{ax + \tan x - \tan 3x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$. If $g'(0)$ exists and is equal to non zero value b , then find $\frac{b}{a}$.

DIBY-L(Graph)

$$f(x) = \begin{cases} \max \{ \sqrt{1+x^2}, \sqrt{4-x^2} \}, & -2 \leq x \leq 0 \\ \min \{ \sqrt{1+x^2}, \sqrt{4-x^2} \}, & 0 < x \leq 2 \end{cases}$$

Discuss continuity and differentiability of $f(x)$.

DIBY-M

Find the number of values of x in $[-4, 4]$ where

$f(x) = [3x + 14] + |4x^2 - 1|(2x^2 + 3x - 2) + \sin\left(\frac{\pi x}{2}\right)$ is non-derivable where $[.]$ is GIF.

DIBY-N

If $f(x) = \begin{cases} |x^5 - 4x^3 + 2| \cos \pi x & x \geq 0 \\ x^3 - \sin(x) - |(x^2 + 8x + 12)x^2| & x < 0 \end{cases}$. Then find number of points where $f(-|x|)$ is not differentiable.

DIBY-O (Graph)

If n_1 = number of points of non-derivability of

$f(x) = \begin{cases} \max. \{|x|, \sin |x|\}, & x \in \left(0, \frac{\pi}{2}\right) \\ \max. \{|x|, \tan |x|\}, & x \in \left(\frac{-\pi}{2}, 0\right) \end{cases}$ in $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ and n_2 = number of points of

discontinuity of $g(x) = [\sin x] + [\cos x]$ where $[.]$ is GIF in $[0, 2\pi]$ then find $(n_1 + n_2)$

RAPID FIRE DIBY's (ANSWERS)



- (A) **Non-differentiable at $x = 1, 2$**
- (B) **-3**
- (C) **0**
- (D) **0**
- (E) **Non-differentiable**
- (F) **Non-differentiable**
- (G) **Non-differentiable**
- (H) **(i) Discontinuous**
(ii) Continuous & differentiable

- (I) **9**
- (J) **2**
- (K) **$\frac{7}{52}$**
- (L) **Non-differentiable at $x = 0, 2, \pm\sqrt{\frac{3}{2}}$**
- (M) **Non-derivable at 25 points**
- (N) **5**
- (O) **6**

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians